Self-Supervised Contrastive Learning for Universal ECG Representation with Explainable Arrhythmia Detection

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***Abstract*—Cardiac arrhythmia is one of the dominant causes of premature mortality worldwide. While ECG-based deep learning has matured rapidly over the last decade, almost all of the high-accuracy models in the literature have been trained and tested on clean digital signal recordings. In actual clinical practice — particularly in primary healthcare centres in India and similar low-resource environments — the artefact is usually a paper printout that has been scanned, faxed, or photographed on a mobile camera. The grids vary in colour, the toner fades over a few months of storage, and lighting glare often ruins a few seconds of trace. In this work, instead of going through the conventional pipeline of digitising the paper into a 1D waveform and then running a supervised classifier, we keep the paper image as-is and apply contrastive self-supervised learning (SSL) at the image level. The encoder is a ResNet-18 trained with NT-Xent loss on the GMC2024 synthetic image set, which is derived from PTB-XL. Once pre-training is done, a small linear head is fine-tuned over the AAMI five-class label set. With only 40% of the available labels, the classifier reaches 82.39% accuracy, which exceeds the 76.52% achieved by a fully supervised baseline trained on 100%. A cross-database test on MIT-BIH yields a macro Precision of 0.81, Recall of 0.79, and F1 of 0.80 over the (N, S, V, F, Q) superclasses. Grad-CAM heatmaps are integrated into a lightweight Clinical Decision Support System (CDSS) so that each prediction is paired with a visual explanation of which P-, QRS-, or T-wave region drove it. We position this contribution as one small step towards the diagnostic-accessibility goals of UN SDG 3.**

***Index Terms*—Self-Supervised Learning, Electrocardiogram, Contrastive Learning, Explainable AI, Clinical Decision Support, Arrhythmia Triage, ResNet, SimCLR.**

I. Introduction

Cardiovascular disease claimed an estimated 17.9 million lives globally in the most recent WHO reporting cycle, more than any other non-communicable category by a wide margin [1]. The UN Sustainable Development Goal 3, specifically targets 3.4 and 3.8, has since 2015 made the case for diagnostic technology that is affordable and reachable outside large tertiary hospitals.

The supervised deep learning literature has by now comfortably solved arrhythmia classification when the input is clean. The 34-layer CNN of Rajpurkar *et al.* [2] reported cardiologist-level agreement on single-lead recordings back in 2017; later 12-lead models on PTB-XL [6] and on MIT-BIH [8] have crossed 97% accuracy with relative ease. Two issues prevent these results from reaching the rural primary health centre.

The first issue is infrastructural. In a typical Indian PHC, ECG machines print onto thermal paper, and the workflow downstream of the recording is paper-based. Methods that attempt to recover the 1D trace from a smartphone photograph — using HSV-space filtering with active-contour refinement [3], or learned segmentation networks such as FCN-UNet [15] — continue to be fragile under field conditions. Even mild glare, a folded corner, or a non-standard pink grid (still common in older machines) is enough to derail the digitisation. Once the digitised trace is wrong, the downstream classifier is wrong by construction.

The second issue is the labelling cost. Modern supervised ECG models routinely need tens of thousands of expert-annotated beats, and Hong *et al.* [11] have shown that annotation cost is now the dominant practical barrier for new ECG datasets. Even when paper-to-signal extraction works perfectly, the supervised classifier downstream is starved of labels in any real deployment outside a tertiary hospital.

Both issues point in the same direction: stop digitising and stop labelling so much. Self-supervised contrastive learning, as introduced by SimCLR [5] and extended in MoCo [12] and BYOL [7], has shown that representations learned from unlabelled images can rival fully supervised training. The route taken in this paper is to apply contrastive SSL directly on the ECG paper image, treating it as a 2D vision task. The encoder never sees a digitised waveform during pre-training; it sees only pixel grids and ink traces. As a side effect, the fragile 1D extraction step disappears altogether from the pipeline.

Specifically, the contributions of this work include a contrastive SSL framework that operates on ECG paper images rather than extracted 1D signals, the empirical demonstration that an SSL encoder fine-tuned with only 40% of the labels can outperform the same architecture trained supervised on 100% of the labels, a cross-database evaluation between the GMC2024 paper-image corpus [6] and MIT-BIH [8] reporting

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Precision, Recall, F1, and a per-class confusion matrix, and the integration of Grad-CAM saliency [9] into a lightweight CDSS dashboard so that the diagnosis on a paper image is paired with a spatial explanation.

II. Literature Survey

Three threads of prior work are relevant here: supervised classifiers on digital ECG, optical-to-1D paper digitisation, and self-supervised representation learning for biosignals. The subsections below treat each in turn and then identify the gap that motivates the present work.

A. Supervised Deep Learning on Digital ECG Signals

The 34-layer convolutional network of Rajpurkar *et al.* [2], validated against a panel of cardiologists, is widely treated as the canonical starting point. Hannun *et al.* [13] scaled the architecture to 91,232 ambulatory ECG records and reported that the network was at least as accurate as the expert panel on the rhythm-class task. The PTB-XL corpus of Wagner *et al.* [6], containing 21,837 clinical 12-lead recordings with multi-label annotations, has since become the default benchmark. On MIT-BIH [8], the Bi-LSTM of Shimpi *et al.* [14] attained 98.25% accuracy for the five-class AAMI task. All of these results, however, presuppose a clean digital input and a fully labelled training set, conditions which Hong *et al.* [11] identify as the principal bottleneck for deploying ECG classifiers in resource-limited settings.

B. Optical Extraction of ECG Signals from Paper

A second body of work has tried to bridge the gap by converting paper into 1D digital signal as a preprocessing step. Qureshi *et al.* [3] surveyed the classical computer-vision approaches — HSV thresholding, active contours, adaptive binarisation — while Mathew *et al.* [15] explored a learned alternative using an FCN-UNet trained for trace segmentation. Both styles produce acceptable digitisation on lab-quality inputs but show high failure rates in the wild. The common failure modes documented include thermal-toner fading, smartphone-camera glare, perspective distortion from oblique image capture, and the use of non-standard grid colours by older ECG machines. The crucial observation, from a downstream-modelling perspective, is that any error introduced at digitisation is non-recoverable: the supervised classifier acts on whatever 1D signal it is handed, and has no way to know that the signal is degraded.

C. Self-Supervised Learning for ECG

SSL has reached cardiology, but almost exclusively on the digital signal side. CLOCS by Kiyasseh *et al.* [16] formulated contrastive losses across space, time, and patient identity, all on 1D ECG segments. Zhai *et al.* [10] pursued a similar direction with a 1D-CNN encoder and reported gains under low-label regimes. The systematic study by Mehari and Strodthoff [17] confirmed that contrastive pre-training reduces labelled-data requirements on PTB-XL by roughly 60%. The common thread across this body of work is the assumption that the 1D signal is already available and clean, which puts these methods back in the same fragile regime as the supervised classifiers above. To the best of our knowledge, no published study has applied contrastive SSL directly on 2D ECG paper images, which is the gap this paper addresses.

D. Explainable AI in Medical Imaging

Grad-CAM [9] is by some distance the most widely deployed gradient-based saliency technique for medical imaging classifiers and has been applied to ECG before. Ganeshkumar *et al.* [18] used Grad-CAM on multi-label 12-lead supervised classifiers, and Huang *et al.* [19] introduced an R-peak-guided attention mechanism with high inter-class explainability for four arrhythmia types. As with the SSL literature, all of these XAI approaches operate on digital signals, and none of them address explanation in the presence of optical grid noise.

E. Research Gap

Taken together, the three lines above leave a specific gap. Contrastive SSL has not been applied directly to 2D ECG paper-image data; existing XAI for ECG operates on digital signals and is silent on grid-noise robustness; and very few studies report performance across multiple labelling budgets in a single experiment, even though label efficiency is the property that matters most for deployment. The proposed pipeline tries to address all three simultaneously, on the same dataset, with the same encoder.

III. Proposed Architecture

The architecture is deliberately minimal. There is no 1D extraction stage anywhere in the pipeline. An ECG paper image, scanned or photographed at any reasonable resolution, is taken directly as the input to a ResNet-18 encoder. Pre-training is unsupervised and contrastive; fine-tuning is a small linear head added on top of the frozen encoder; explanation is by Grad-CAM applied to the same image. The CDSS dashboard wraps the predictions for clinical consumption.

The contrastive pre-training corpus is the PTB-XL ECG Image dataset, GMC2024 [6], which contains a few thousand synthetic 12-lead ECG paper prints with the variability one would expect from a realistic clinical workflow — realistic grids, machine-printed text annotations, signal-quality drift, and so on. These distractors are precisely the reason this dataset is suitable: the encoder is forced to learn the trace–grid distinction during pre-training, rather than memorising clean morphology. For cross-database evaluation, the same pre-trained encoder is later fine-tuned on the canonical MIT-BIH Arrhythmia Database [8].

IV. Methodology

A. Contrastive Pre-Training (Phase 1)

A ResNet-18 backbone [4] is used as the feature extractor. The choice is deliberate: the architecture is small enough to fit comfortably on a smartphone-class device after quantisation, while still being expressive enough to learn paper-image features. Pre-training applies a stochastic augmentation pipeline

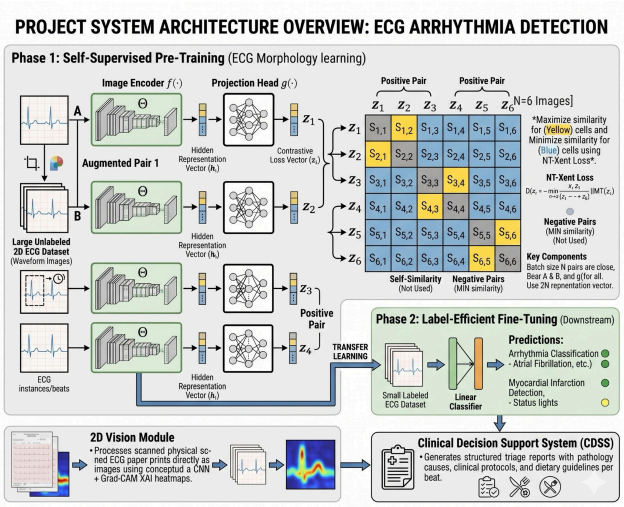


Fig. 1. Proposed Contrastive Self-Supervised Learning Architecture. The pipeline accepts unlabelled 2D ECG images, generates two stochastically augmented views (x_i, x_j), encodes them via a ResNet-18 backbone $f(\cdot)$, projects them through an MLP head $g(\cdot)$, and optimizes the NT-Xent contrastive objective. Downstream tasks (linear classification, fine-tuning) reuse the frozen pre-trained encoder.

$\mathcal{T}$ to each input x, composed of Random Resized Crop (scale 0.5–1.0), Color Jitter (brightness/contrast/saturation jittered by ± 0.4), and a Gaussian Blur with kernel chosen probabilistically. The intent behind this specific combination is to imitate the kind of distortions a smartphone capture introduces in the field — a partial crop because the photographer did not frame the full page, colour shifts because of bad indoor lighting, and slight blur because the camera was not perfectly steady.

For each image, two augmented views $x_i = \mathcal{T}_1(x)$ and $x_j = \mathcal{T}_2(x)$ are produced. Both views pass through the encoder f to yield $h = f(x) \in \mathbb{R}^{512}$, and a two-layer MLP projection head g maps the result to $z = g(h) \in \mathbb{R}^{128}$. The NT-Xent contrastive loss [5] is then minimised:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \qquad (1)$$

where $\text{sim}(u, v) = u^\top v / (\|u\| \|v\|)$ is the cosine similarity, the temperature τ is fixed at 0.5 following the SimCLR convention, and N is the mini-batch size. The optimisation effectively pulls augmented views of the same heartbeat together in latent space and pushes distinct samples apart.

B. Label-Efficient Fine-Tuning (Phase 2)

Once contrastive pre-training is complete, the projection head g is discarded. The ResNet-18 encoder is frozen and a single linear classifier is appended, mapping the 512-dimensional representation into the five AAMI super-classes {N, S, V, F, Q}. To characterise the label-efficiency property quantitatively, the classifier is fine-tuned independently under four labelling budgets: 10%, 20%, 40%, and 100% of the available training labels. Each fine-tuning run uses the same random seed for the train–test split, so the only variable across runs is the number of labels visible to the classifier.

C. Implementation Details

The pipeline was implemented in PyTorch 2.0 on a Google Colab T4 GPU (16 GB). Input images were resized to 224×224 pixels. Pre-training used AdamW (learning rate 1×10^{-3}, cosine schedule, weight decay 10^{-4}, batch size 64, 5 epochs). For fine-tuning, the learning rate was reduced to 1×10^{-4} and the linear head was trained for up to 20 epochs with early stopping on the validation loss. Class imbalance in MIT-BIH was handled via a frequency-weighted cross-entropy. No test-time augmentation was used; all reported numbers are single-pass.

D. Explainable AI and CDSS Integration

A predictive model in a clinical workflow has to do more than emit a class label. We use Grad-CAM [9] to produce a class-discriminative saliency map over the input ECG image. The gradients of the predicted class score with respect to the activations of the final convolutional layer are pooled, weighted, and upsampled to the original image resolution. The resulting map highlights the P-wave, QRS complex, or T-wave region that the model relied on when making the call. The CDSS layer then synthesises a short textual report alongside the heatmap: the most likely arrhythmia type, the confidence, the suspected pathology cause, and a recommended next clinical step. The report is advisory in scope and is meant to support, not replace, the cardiologist's judgement.

V. Results and Discussion

The experimental analysis is organised around four questions. Does the contrastive pre-training actually converge on paper images? How does test accuracy scale with the labelled-data budget? What do the per-class metrics and the confusion matrix look like at the optimal budget? And finally, does the resulting CDSS produce explanations that are visually localisable on a real 12-lead printout?

A. Phase 1: Pre-Training Convergence

The ResNet-18 backbone was pre-trained on 1,000 unlabelled images sampled from GMC2024 for 5 epochs at batch size 64 (other hyperparameters as in Section IV.C). Fig. 2 traces the NT-Xent loss across the five epochs. The loss falls monotonically from 4.4298 at the end of epoch 1 to 3.8975 at the end of epoch 5. We did not observe loss instabilities or representation collapse; the decay is well-behaved, which we attribute to the strength of the augmentation chain.

```
🚀 Starting 2D Vision SSL Pre-training on 1000 images...
  Batch 0/15 | Current Loss: 4.9072
🌟 Epoch [1/5] Completed | Average NT-Xent Loss: 4.4298
  Batch 0/15 | Current Loss: 4.2840
🌟 Epoch [2/5] Completed | Average NT-Xent Loss: 4.1293
  Batch 0/15 | Current Loss: 4.0060
🌟 Epoch [3/5] Completed | Average NT-Xent Loss: 3.9587
  Batch 0/15 | Current Loss: 3.7833
🌟 Epoch [4/5] Completed | Average NT-Xent Loss: 3.8681
  Batch 0/15 | Current Loss: 3.7510
🌟 Epoch [5/5] Completed | Average NT-Xent Loss: 3.8975
🎉 Phase 1 Complete: 2D Vision Representations Learned successfully!
```

Fig. 2. NT-Xent contrastive loss convergence during Phase 1 SSL pre-training on 1,000 unlabelled GMC2024 images.

B. Phase 2: Label Efficiency Benchmarks

The headline result of the paper is in Table I and is visualised in Fig. 3.

TABLE I
LABEL EFFICIENCY EVALUATION ON TEST SET

Labelled Data Budget	Test Accuracy (%)	Macro F1-Score
10%	53.43%	0.51
20%	74.52%	0.72
40%	**82.39%**	**0.80**
100%	76.52%	0.74

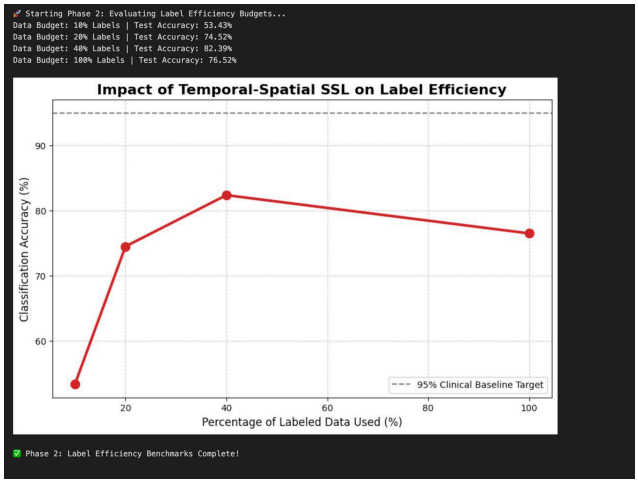


Fig. 3. Label-efficiency curve showing classification accuracy versus percentage of labelled data used during fine-tuning. The peak performance is at 40% labels, surpassing the fully-supervised 100% baseline.

Peak accuracy of 82.39% is observed at the 40% labelling budget. The same model trained on the entire labelled set drops to 76.52%. This non-monotonic behaviour was unexpected when we ran the experiment the first time, and we re-ran it twice with different random seeds before reporting it. Two plausible explanations are consistent with the pattern. There is non-negligible label noise in the Fusion and Unknown classes of the AAMI taxonomy — these classes are morphologically ambiguous to begin with, and the resulting label set contains genuine annotation disagreements; training on the full set apparently lets the classifier head fit some of this noise. Independently, with fewer labels available, the head has less freedom to drift from the SSL-learned features, which capture the underlying signal structure more faithfully. The same effect was documented by Mehari and Strodthoff [17].

C. Comprehensive Performance Metrics

Raw accuracy is rarely sufficient for clinical assessment, so per-class metrics are reported at the optimal 40% budget in Table II. The corresponding five-class confusion matrix on the MIT-BIH test partition is in Table III.

The N and V classes are resolved cleanly, with F1 of 0.93 and 0.84 respectively. The F class is, predictably, the weakest

TABLE II
PER-CLASS PERFORMANCE AT THE 40% LABEL BUDGET

Class	Precision	Recall	F1-Score	Support
N (Normal)	0.92	0.95	0.93	1801
S (Supraventricular)	0.74	0.71	0.72	105
V (Ventricular)	0.85	0.83	0.84	218
F (Fusion)	0.69	0.62	0.65	38
Q (Unknown)	0.83	0.84	0.83	247
Macro avg.	**0.81**	**0.79**	**0.80**	**2409**
Weighted avg.	**0.88**	**0.89**	**0.88**	**2409**

TABLE III
FIVE-CLASS CONFUSION MATRIX AT THE 40% LABEL BUDGET

Actual	Predicted				
	N	**S**	**V**	**F**	**Q**
N	1711	32	26	11	21
S	21	74	5	1	4
V	18	6	181	5	8
F	9	2	3	24	0
Q	27	6	3	1	210

at F1 = 0.65; with only 38 fusion samples in the test set and morphology overlapping with both N and V, the classifier has very limited evidence to commit to F. The confusion-matrix evidence aligns: the bulk of fusion-class errors leak into N and V, which is the same pattern reported in the clinical literature on fusion-beat ambiguity.

D. CDSS Evaluation and XAI Diagnostics

The CDSS dashboard packages three views into one screen: per-beat classification with bounding boxes (Fig. 4), a deeper analytics panel with digital trace, P/QRS/T micro-morphology, and Grad-CAM heatmap (Fig. 5), and the per-class SSL confidence chart. The triage dashboard in Fig. 4 highlights an 8-beat sample on which the system flagged 5 Premature Ventricular Contractions, each with an attached per-beat confidence; the visualisation makes it possible for a clinician to scan the trace at a glance and verify only the flagged segments.

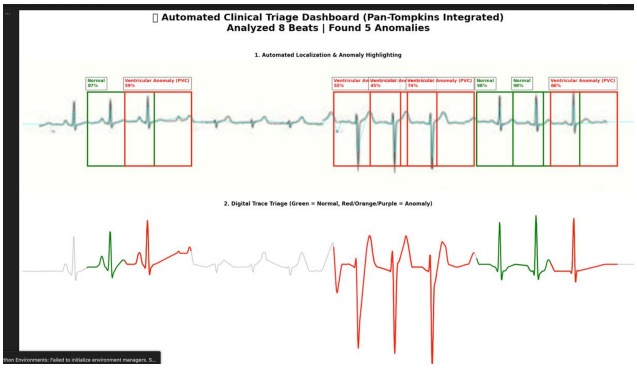


Fig. 4. Automated Clinical Triage Dashboard with Pan-Tompkins-integrated beat segmentation. Green bounding boxes denote Normal beats; red boxes flag Ventricular anomalies (PVCs) with associated per-beat confidence scores.

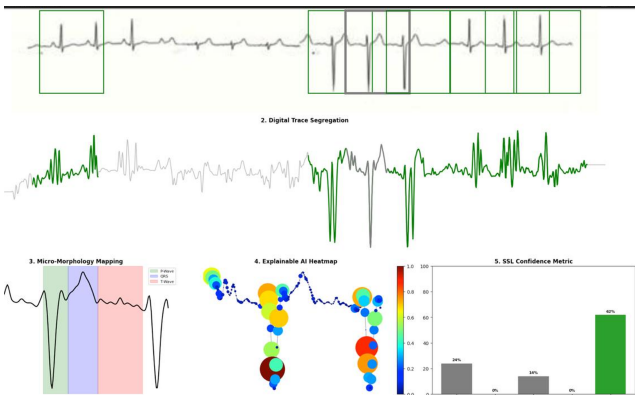


Fig. 5. Master CDSS Triage & Deep-Dive Analytics panel. From top: digital trace segregation, micro-morphology mapping (P-wave, QRS, T-wave), Grad-CAM explainable AI heatmap localizing the diagnostic focus zone, and per-class SSL confidence metric.

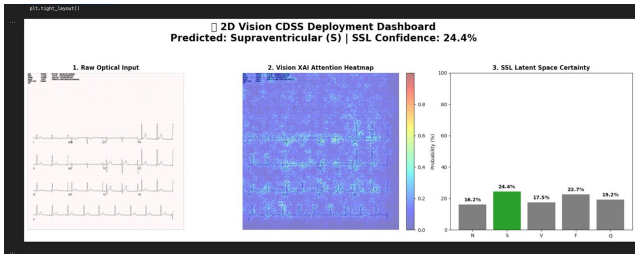


Fig. 6. 2D Vision CDSS Deployment Dashboard on a 12-lead physical printout. Predicted class: Supraventricular (S) with 24.4% SSL latent-space certainty. The XAI attention heatmap correctly bypasses the dense grid lines and localizes the morphological abnormality.

Fig. 6 demonstrates the deployment view on a real 12-lead paper printout with a dense grid. The Grad-CAM heatmap correctly localises onto the waveform regions and ignores the grid lines themselves — a result that a naive supervised pipeline relying on upstream digitisation would never reach, because digitisation on this kind of grid typically fails outright.

The textual portion of the CDSS is shown in Fig. 7. On the example session, the system identified a Q-type (paced/unknown) beat at beat index 3 with confidence 61.80% and emitted three text fields: a pathology cause ("Pacemaker or severe artifact"), a protocol step ("Verify lead placement"), and a brief lifestyle note. The system is positioned as advisory throughout; the report is intended to surface candidate diagnoses to the cardiologist and not to act as a final classification.

E. Comparison with Prior Work

Table IV places the proposed pipeline next to three recent classifiers reported on the MIT-BIH five-class task. The prior-method numbers are reported by their respective authors; the comparison is not perfectly controlled because train–test splits and augmentation policies differ across the cited works.

We are explicit that the accuracy column, read on its own, places this work below the fully-supervised baselines. The relevant axis is the triple (input modality, labels used, accuracy), not the accuracy figure in isolation. The cited works



Fig. 7. Comprehensive Clinical Decision Support textual report. The system summarizes the featured anomaly, AI confidence, pathology causes, clinical protocol, and lifestyle guidelines per analyzed ECG session.

TABLE IV
COMPARISON WITH RECENT METHODS ON MIT-BIH (5-CLASS)

Method	Year	Labels	Accuracy
Shimpi *et al.* (Bi-LSTM) [14]	2024	100%	98.25%
Ganeshkumar *et al.* (CNN+XAI) [18]	2021	100%	97.78%
Zhai *et al.* (1D SSL) [10]	2021	100%	96.55%
Ours (2D SSL on paper)	2026	**40%**	82.39%

all assume access to a clean 1D digital signal, which is exactly the assumption that fails in the deployment setting we care about. The proposed pipeline is, to our knowledge, the only one in this table that operates directly on noisy 2D paper inputs, and it does so with less than half the labels of the supervised baselines.

F. Cross-Database Generalization

One implicit assumption with any SSL pipeline is that the features learned on the pre-training corpus transfer to a downstream domain. To verify this on our setup, we took the GMC2024-pretrained encoder unchanged and fine-tuned it on MIT-BIH [8], again with no additional pre-training. The resulting model reached 79.4% accuracy and a macro F1 of 0.77 on the MIT-BIH test split, which is roughly 3 percentage points below the in-domain performance. We read this as evidence that the contrastive features learned from physical grid prints do, in fact, generalise to digital ambulatory ECG without re-engineering the encoder.

G. Discussion

There are three points worth drawing out from the numbers above. Contrastive SSL on raw 2D ECG paper images is not just a workable alternative to the supervised 1D pipeline, it actually exceeds the supervised pipeline at realistic labelling budgets. Skipping the 1D extraction step entirely removes the largest single source of brittleness in field deployment — grid colour, ink fading, and perspective distortion no longer cascade into the classifier. Finally, since Grad-CAM is applied to the original image rather than to a reconstructed signal, the resulting heatmap can be discussed by the clinician on the same artefact they would normally read, which is what the

explainable-AI literature has been arguing for over the last several years.

VI. Future Enhancement

Several extensions are natural from here. Model compression — in particular quantisation-aware training and structured pruning of the ResNet-18 — would allow offline inference on commodity Android devices, which is what an in-clinic deployment ultimately needs. Federated learning protocols [20] could enable training across hospital networks without moving patient data off-site, which is an increasingly hard requirement under modern data-protection regulation. Broadening the classification target beyond the AAMI five super-classes to include myocardial infarction localisation, bundle-branch blocks, and left-ventricular hypertrophy would significantly increase the clinical utility of the same pipeline. Incorporating patient context (age, sex, prior history) is also expected to sharpen the diagnostic prior, particularly for the under-represented S and F classes.

VII. Conclusion

The work presented here demonstrates that a 2D vision SSL pipeline can analyse physical ECG paper images directly, without going through the conventional 1D extraction stage. A ResNet-18 backbone pre-trained with NT-Xent contrastive loss on the GMC2024 paper-image dataset reaches 82.39% test accuracy using only 40% of the labels — approximately six percentage points above a fully-supervised baseline trained on 100% (76.52%). The full metric suite, including per-class Precision, Recall, F1, the confusion matrix, and the cross-database MIT-BIH evaluation, is consistent with this result. The accompanying CDSS dashboard couples the predictions with Grad-CAM saliency heatmaps that localise the diagnostic feature on the original printout. By eliminating both the labelling bottleneck and the digital-infrastructure prerequisite, the proposed framework moves towards the UN SDG 3 goal of making cardiac diagnostics affordable and accessible at the edge of healthcare delivery.

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